WORLDALIGN: DECOUPLED 4D REWARD FOR WORLD-CONSISTENT VIDEO GENERATION

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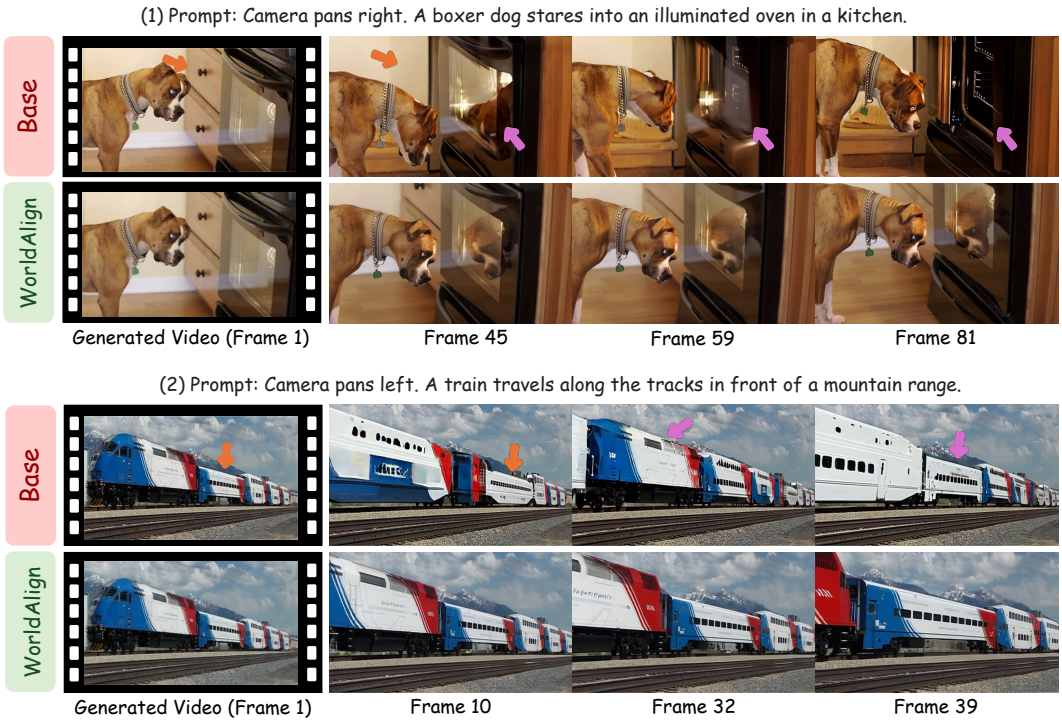


Figure 1: Comparison between WorldAlign and the base model (Wan2.1-I2V-14B-480P). The first case highlights improvements in static consistency: the base model loses the background cabinetry (orange arrows), while the oven handle disappears and its door deforms before vanishing (magenta arrows). The second case highlights improvements in dynamic consistency: the train cars abruptly change color and shape and move opposite to the locomotive's heading (see the cars marked by same-colored arrows). WorldAlign instead preserves static scene structure and plausible dynamic evolution. If the videos fail to play, **we strongly recommend using a multimedia-enabled PDF reader**. See the supplementary material for more examples.

ABSTRACT

Faithful visual world simulation requires generated videos to maintain 4D world consistency, encompassing both static and dynamic consistency. Static consistency requires coherent 3D structure in static environments across viewpoints, while dynamic consistency requires plausible subject motion and consistent appearance over time. Geometry-aware post-training offers a promising way to improve world consistency. However, existing methods often rely on a static-scene assumption. Even those that accommodate dynamic scenes struggle to provide reliable static-consistency feedback, while dynamic consistency is often overlooked or inadequately assessed. To address these limitations, we introduce *WorldAlign*, a decoupled 4D reward framework that semantically separates static regions and dynamic subjects and provides feedback by aligning each with a world prior suited to its assumptions. For static regions, WorldAlign aligns static geometry with a geometric world prior through semantically guided masked reprojection, en-

abling more reliable static-consistency evaluation; an auxiliary camera-motion reward discourages nearly static solutions. For dynamic subjects, WorldAlign uses a strong vision-language model (VLM) as a dynamic world prior and constructs a VLM-as-a-judge reward based on sample-specific checklists that assess dynamicity, physical plausibility, shape, and texture consistency. This decoupled design enables more effective online post-training without requiring human preference annotations. Across two pretrained image-to-video generators, Wan2.1 and Wan2.2, WorldAlign jointly improves static and dynamic consistency over existing methods without suppressing overall or subject motion. These results support decoupled world-prior alignment for more faithful visual world simulation.

1 INTRODUCTION

Faithful visual world simulation requires more than frame-wise realism: the generated world must remain coherent over time. We refer to this property as world consistency, which comprises static and dynamic consistency. Static consistency requires static environments to maintain coherent 3D structure across viewpoints. Dynamic consistency requires subjects to move plausibly while preserving their appearance over time. Such consistency is essential when video generation models serve as world simulators for robotics (Shan et al., 2026; Zhang et al., 2026), autonomous driving (Hu et al., 2023; Wang et al., 2024; Gao et al., 2024), and interactive games (Bruce et al., 2024; Sun et al., 2025; Team et al., 2026). Despite substantial advances in high-fidelity, open-domain video synthesis (Brooks et al., 2024; Seedance et al., 2026; Team et al., 2025; Kong et al., 2024; Wan et al., 2025), large-scale pretrained video generation models remain insufficient for faithful world simulation. A key reason is that their pretraining objectives primarily optimize prediction in RGB or latent space, leaving 3D structure and physical dynamics to be learned only implicitly, without explicit world-consistency constraints. Consequently, static structures may slide, warp, or flicker across viewpoints, while dynamic subjects may exhibit implausible motion and appearance changes, as shown in Fig. 1. Achieving world consistency therefore remains a central challenge for faithful visual world simulation.

Existing efforts toward world-consistent video generation broadly follow two routes. The first modifies the generator architecture to integrate explicit geometric information, including point-conditioned generation (Ren et al., 2025; Yu et al., 2024; Liu et al., 2026a; Zhao et al., 2026) and joint RGB-geometry modeling (Zhang et al., 2025; Bai et al., 2026; Dai et al., 2025; Chen et al., 2025). Although effective, the additional modules and geometric supervision may compromise the broad generalization inherited from large-scale video pretraining. The second preserves the pretrained generator and improves consistency through post-training alignment, allowing expert knowledge or world priors to provide feedback while retaining its broad generative capability. However, real-world scenes often contain both static environments and dynamic subjects, making reliable consistency feedback in such scenes a key challenge for post-training. Epipolar-DPO, VideoGPA, and the geometric rewards of World-R1 rely on static-scene assumptions (Kupyn et al., 2025; Du et al., 2026; Wang et al., 2026). In dynamic scenes, independently moving subjects violate these assumptions, causing legitimate subject motion to be penalized and potentially driving the generator toward frozen subjects. VGGRPO and GeoFlow accommodate dynamic scenes by using motion cues to exclude or downweight dynamic regions during static-consistency evaluation (An et al., 2026; Ackermann et al., 2026). However, alongside legitimate subject motion, these cues also identify unintended sliding or warping of static regions as dynamic, weakening the static-consistency signal (see Fig. 4). Dynamic consistency is also insufficiently addressed: VGGRPO lacks dedicated feedback for dynamic subjects, while GeoFlow focuses on appearance consistency without explicitly assessing whether subjects move as intended or whether their motion is physically plausible. These limitations motivate us to develop more reliable and comprehensive world-consistency rewards.

To this end, we introduce *WorldAlign* (Fig. 2), a decoupled 4D reward framework that semantically separates static regions and dynamic subjects and provides feedback by aligning each with a world prior suited to its assumptions, rather than imposing a static-scene assumption on the entire scene. For static consistency, WorldAlign first semantically segments and tracks dynamic subjects rather than relying on motion cues. This enables geometric evaluation to exclude independently moving subjects while retaining static regions, including those exhibiting unintended sliding or warping. WorldAlign then uses a geometry foundation model (Wang et al., 2025) as a geometric world prior to

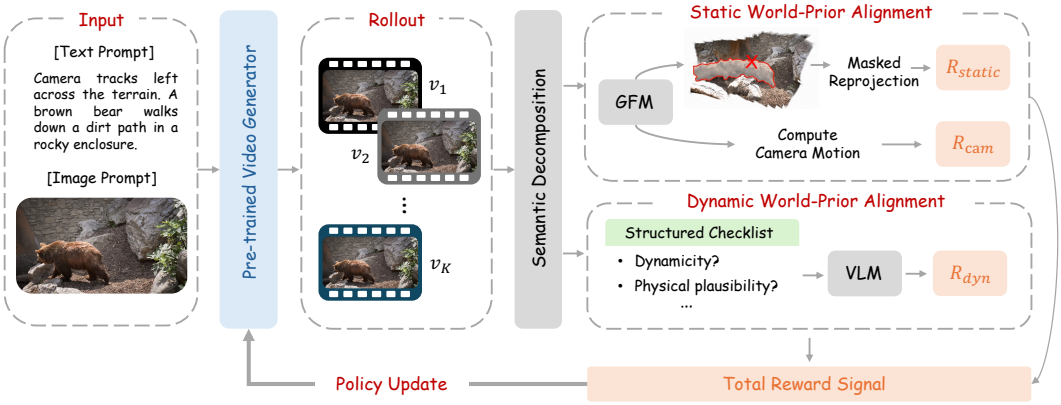


Figure 2: Overview of the proposed WorldAlign. Given generated videos, WorldAlign semantically segments and tracks dynamic subjects to decouple each video into static and dynamic regions. Static regions are aligned with a geometric world prior through masked reprojection and camera-motion rewards, while dynamic subjects are aligned with a VLM-based world prior through a structured *VLM-as-a-judge* reward. The reward signals are normalized within each same-prompt sample group and aggregated for online policy update.

construct a shared point cloud and reproject it into the input views. Dynamic subjects are masked out during both point-cloud construction and reprojection-error computation, reducing their interference with static-consistency evaluation. When the frames depict a coherent static 3D scene, the shared geometry should reproduce their static regions consistently, whereas inconsistent structures lead to larger reprojection errors. The resulting masked reprojection reward therefore penalizes inconsistent static structures without penalizing legitimate subject motion. Since reducing camera motion can make geometric consistency easier to satisfy, an auxiliary camera-motion reward discourages nearly static solutions and encourages meaningful viewpoint changes. For dynamic consistency, WorldAlign explicitly evaluates their motion and appearance over time. Because plausible subject evolution depends on the subject's properties and scene context, we use a strong vision-language model (VLM) as a dynamic world prior. We construct a VLM-as-a-judge reward based on sample-specific checklists that translate this world knowledge into explicit criteria covering dynamicity, physical plausibility, shape consistency, and texture consistency. The fraction of satisfied checklist items provides dynamic-consistency feedback that evaluates not only appearance preservation but also whether subjects move as intended and whether their motion is physically plausible. Together, these rewards enable online post-training to jointly improve static and dynamic consistency without modifying the generator architecture or requiring human preference annotations.

To validate WorldAlign, we compare it with state-of-the-art post-training methods across two pre-trained image-to-video generators using both reward-related and reward-independent metrics. Experiments demonstrate that WorldAlign improves static geometry and dynamic consistency over existing methods without suppressing camera or subject motion. These results support decoupled world-prior alignment as an effective approach to more faithful visual world simulation.

2 Related Work

2.1 Pretrained Video Generation Models

Large-scale pretrained video generation models (Brooks et al., 2024; Seedance et al., 2026; Team et al., 2025; Kong et al., 2024; Wan et al., 2025) have substantially advanced open-domain video synthesis. Their progress is driven by diffusion (Ho et al., 2020) and flow-matching objectives (Lipman et al., 2022; Liu et al., 2022), transformer-based architectures (Dosovitskiy et al., 2020), and large-scale video-text training. However, their pretraining objectives primarily optimize prediction in RGB or latent space, leaving 3D structure and physical dynamics to be learned only implicitly, without explicit world-consistency constraints. Consequently, world consistency remains unreli-

able across generations. WorldAlign therefore explicitly optimizes the world consistency of these pretrained models through post-training.

2.2 World-Consistent Video Generation

Existing approaches improve world consistency through two main routes: ① **architecture modification for geometry integration**, which modifies the generator architecture to incorporate geometric information into the generation process, and ② **geometry-aware post-training**, which derives geometric feedback from generated videos to refine pretrained generators through post-training.

Architecture Modification for Geometry Integration. Methods in this route introduce additional conditioning inputs derived from reconstructed 3D points and train the generator to use them (Ren et al., 2025; Yu et al., 2024; Liu et al., 2026a; Zhao et al., 2026). Other methods extend the generator to jointly predict RGB appearance and geometric representations such as depth or 3D coordinates (Zhang et al., 2025; Bai et al., 2026; Dai et al., 2025; Chen et al., 2025). These designs require architectural modifications and additional training to accommodate geometric inputs or outputs, which may compromise the broad generalization acquired through large-scale pretraining. WorldAlign instead improves the world consistency through post-training alignment, leaving the generator architecture unchanged.

Geometry-Aware Post-Training. Epipolar-DPO and VideoGPA construct preference pairs from generated videos for post-training, with preferences determined by epipolar error in Epipolar-DPO (Kupyn et al., 2025) and reprojection errors computed from predicted geometry and camera poses in VideoGPA (Du et al., 2026). Both geometric criteria rely on static-scene assumptions; applying them directly to dynamic scenes can penalize legitimate subject motion and encourage the generator to produce stationary subjects. World-R1 (Wang et al., 2026) instead guides post-training with a 3D-aware reward alongside a general visual reward, but this 3D-aware reward still relies on static-scene assumptions. To accommodate dynamic content, it periodically disables the 3D-aware reward and optimizes dynamic scenes using only the general visual reward, which provides no explicit world-consistency constraints. VGGRPO and GeoFlow can provide geometric feedback for dynamic scenes. They use motion cues to identify dynamic regions and reduce interference from subject motion during static-consistency evaluation. Specifically, VGGRPO (An et al., 2026) uses scene flow to filter out dynamic regions, while GeoFlow (Ackermann et al., 2026) uses optical-flow residuals to downweight their contributions. However, these cues can also identify unintended sliding or warping of static regions as dynamic, weakening the static-consistency signal by excluding or downweighting regions whose errors should be penalized. Dynamic consistency also remains insufficiently addressed: VGGRPO provides no dedicated consistency feedback for the filtered subjects, while GeoFlow measures DINO-based appearance consistency across the whole scene after flow alignment. This appearance-based evaluation does not explicitly assess whether subjects move as intended or whether their motion is physically plausible. To improve world consistency in real-world dynamic scenes, WorldAlign uses tracked semantic masks to exclude independently moving subjects from static-consistency evaluation while retaining inconsistent static regions, providing more reliable static-consistency feedback than methods based on motion cues. It also provides dedicated feedback for dynamic subjects through sample-specific checklists that assess dynamicity, physical plausibility, shape consistency, and texture consistency.

3 Method

3.1 Method Overview

We consider image-to-video generation[1], where a pretrained generator produces an N-frame video $v = \{I_n\}_{n=1}^{N}$ conditioned on an image I^c and a text prompt y. Our goal is to improve the world consistency of generated videos through post-training without modifying the generator architecture.

Real-world scenes contain both static environments and dynamic subjects, which satisfy different consistency assumptions. WorldAlign therefore uses suitable world priors to define separate rewards

[1]WorldAlign can be extended to text-to-video generation by obtaining subject masks from generated videos and constructing checklists from text prompts.

for static regions and dynamic subjects, as illustrated in Fig. 2. Given a generated video v, we initialize a segmentation tracker Carion et al. (2025) with the subject mask M^c of the conditioning image. The tracker propagates the mask across the video to obtain subject masks $M = \{M_n\}_{n=1}^N$, which define the dynamic regions, while their complements $\overline{M}_n$ define the static regions. For static regions, a geometry foundation model is used as the prior for a masked reprojection reward R_{static}, while an auxiliary camera-motion reward R_{cam} discourages nearly static solutions. For dynamic subjects, a vision-language model is used as the prior for a structured *VLM-as-a-judge* reward R_{dyn}, which evaluates motion and appearance consistency using sample-specific checklists. These rewards jointly guide online post-training through GRPO to update the generator.

3.2 Static World-Prior Alignment

Masked Reprojection Reward. Static regions approximately satisfy rigid multi-view geometry. We therefore use a geometry foundation model (GFM) (Wang et al., 2025) to measure their consistency through cross-view reprojection. Given a generated video v, the GFM predicts a pixel-aligned 3D point map P_n, camera intrinsics K_n, and camera extrinsics C_n for each frame I_n.

However, directly aggregating the predicted point maps also fuses the dynamic subject at its successive positions into the shared point cloud, forming a 3D trace of its motion. Directly reprojecting this point cloud into each frame produces motion-induced artifacts. Pixels affected by these artifacts yield large reprojection errors regardless of whether the corresponding static regions are geometrically consistent, making direct reprojection unreliable for static-consistency evaluation, as shown in Fig. 3.

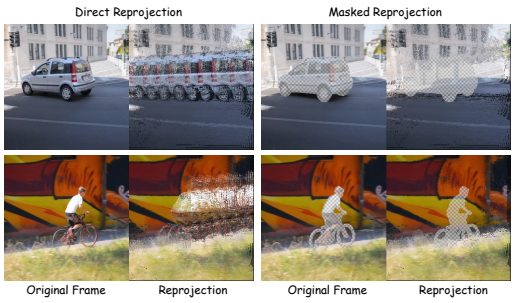


Figure 3: Motion-induced artifacts of Direct Reprojection. Our masked reprojection can isolate static geometry from subject motion, yielding a more reliable static-consistency reward.

To remove this interference, WorldAlign constructs the shared point cloud using only static regions. Because P_n is pixel-aligned, the tracked mask M_n can be applied directly to exclude 3D points corresponding to the dynamic subject:

$$\mathcal{P}_{\text{static}} = \left\{ P_n(x) \mid x \in \overline{M}_n,\ n = 1, \ldots, N \right\}. \tag{1}$$

The resulting point cloud excludes the subject's motion trajectory and retains only the 3D points assigned to static regions. Because dynamic regions are defined by tracked subject masks rather than motion cues, undesired distortions in static regions still remain during static-consistency evaluation. WorldAlign then projects $\mathcal{P}_{\text{static}}$ into each frame using the predicted camera parameters: $\hat{I}_n = \Pi\left(\mathcal{P}_{\text{static}}; K_n, C_n\right)$, where Π denotes the reprojection operation. Although the reprojected frames are rendered from the point cloud containing only the static environments, the original frames still contain dynamic subjects. We therefore apply the same subject mask to each original frame and its reprojected counterpart, excluding the corresponding pixel locations from comparison. We further restrict the comparison to pixels with valid reprojections. The valid comparison set is defined as

$$\Omega = \left\{ (n, x) \mid x \in \overline{M}_n,\ \hat{I}_n(x) \text{ is valid} \right\}. \tag{2}$$

Here, n indexes frames and x denotes a pixel location. A pair (n, x) belongs to Ω only if x lies outside the subject mask in frame n and has a valid value in the corresponding reprojected frame. The masked reprojection error and its corresponding static-consistency reward are

$$E_{\text{rep}} = \frac{1}{|\Omega|} \sum_{(n,x)\in\Omega} \left\| I_n(x) - \hat{I}_n(x) \right\|_2^2, \quad R_{\text{static}} = -E_{\text{rep}}. \tag{3}$$

Camera-Motion Reward. The masked reprojection reward admits a trivial solution: suppressing viewpoint changes can reduce the reprojection error without requiring consistent 3D structure. To

prevent this collapse, WorldAlign reuses the camera extrinsics predicted by the GFM to construct an auxiliary reward that penalizes insufficient camera motion.

Let $C_n = (R_n, t_n)$ denote the predicted camera extrinsics, with rotation R_n and translation t_n. We define the camera-motion magnitude as $m(C_{1:N}) = \bar{d}_{\text{trans}} + \bar{d}_{\text{rot}}$, where $\bar{d}_{\text{trans}}$ and $\bar{d}_{\text{rot}}$ denote the mean adjacent-frame translation $\|t_{n+1} - t_n\|_2$ and mean adjacent-frame rotation angle $\theta(R_{n+1}R_n^\top)$, respectively. The camera-motion reward applies a hinge penalty when the predicted motion falls below a threshold τ:

$$R_{\text{cam}} = -\max\left(0, \tau - m(C_{1:N})\right), \tag{4}$$

where τ is the lower-bound camera-motion threshold. This reward penalizes videos with insufficient viewpoint changes, preventing the generator from reducing the reprojection error solely through nearly static viewpoints.

3.3 Dynamic World-Prior Alignment

WorldAlign uses a strong VLM as a dynamic world prior and translates its world knowledge into a structured *VLM-as-a-judge* reward through subject-specific binary questions.

Before training, we use a VLM to construct a sample-specific checklist $\mathcal{Q} = \{q_j\}_{j=1}^{J}$ from each conditioning pair (I^c, y). The checklist consists of subject-centric binary questions tailored to the subject and its intended motion, covering four complementary aspects of dynamic consistency:

- **Dynamicity**: whether the subject exhibits the intended motion rather than remaining static.
- **Physical plausibility**: whether the motion follows physically plausible dynamics given the subject's intrinsic properties and the surrounding environment.
- **Shape**: whether the subject preserves its structure over time while allowing physically plausible articulation.
- **Texture**: whether the subject preserves its defining visual attributes, such as color, patterns, and surface details.

Together, the shape and texture criteria prevent appearance drift that could make the subject appear as a different entity. The sample-specific checklist accounts for subject- and context-dependent dynamics, while its binary format decomposes an open-ended judgment into explicit criteria for more reliable evaluation.

During training, the judging VLM receives the full generated video v and the checklist $\mathcal{Q}$. The full video allows it to consider the surrounding scene, while the evaluation instruction directs it to focus on the dynamic subject and answer each question with YES or NO. Let $a_j \in \{0, 1\}$ denote the answer to question q_j, with YES mapped to 1. We define the *VLM-as-a-judge* reward as

$$R_{\text{dyn}} = \frac{1}{J}\sum_{j=1}^{J} a_j. \tag{5}$$

Thus, R_{dyn} measures the fraction of checklist criteria satisfied by the generated video.

3.4 Online Policy Update

We optimize the pretrained generator using online GRPO following Flow-GRPO (Liu et al., 2026b). For each conditioning pair (I^c, y), the generator samples a group of K videos $\{v_i\}_{i=1}^{K}$. For every v_i, WorldAlign computes the three reward components R_{static}, R_{cam}, and R_{dyn}.

To obtain group-relative signals while balancing their different scales, we normalize each reward component separately within the group:

$$\widehat{R}_k(v_i) = \frac{R_k(v_i) - \mu_k}{\sigma_k + \epsilon}, \qquad k \in \{\text{static}, \text{cam}, \text{dyn}\}, \tag{6}$$

where μ_k and σ_k are the group mean and standard deviation of reward k, and ϵ is a small constant for numerical stability. The normalized components are aggregated as the advantage:

$$A_i = \lambda_s \widehat{R}_{\text{static}}(v_i) + \lambda_c \widehat{R}_{\text{cam}}(v_i) + \lambda_d \widehat{R}_{\text{dyn}}(v_i), \tag{7}$$

where λ_s, λ_c, and λ_d control the contributions of the three reward components. The resulting advantage is used with the standard clipped GRPO objective to update the generator. The tracker, GFM, and VLM remain frozen and are used only for reward computation.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Datasets. WorldAlign focuses on real-world scenes in which static environments and dynamic subjects coexist. Existing geometry-alignment datasets, such as DL3DV (Ling et al., 2024), contain only static scenes and therefore cannot support dynamic consistency alignment. We thus construct training and evaluation data from DAVIS (Pont-Tuset et al., 2017) and COCO (Lin et al., 2014). Each sample comprises a conditioning image, a subject mask for the conditioning image, an image-to-video prompt, and a sample-specific dynamic-consistency checklist. The resulting dataset contains 1,389 samples, of which 100 are randomly held out for evaluation and the remaining 1,289 are used for training. Further details on data selection, prompt generation, and checklist construction are provided in the supplementary material.

Baselines. We evaluate WorldAlign on Wan2.2-TI2V-5B and Wan2.1-I2V-14B-480P (Wan et al., 2025), using each pretrained generator as a base model, and compare against Epipolar-DPO (Kupyn et al., 2025), VideoGPA (Du et al., 2026), World-R1 (Wang et al., 2026), and GeoFlow (Ackermann et al., 2026). For a controlled comparison, all methods are trained on the same data with the same base generators and matched generation settings. Our World-R1 baseline is a reproduction adapted to the trajectory-free setting shared by all methods. It retains the reconstruction, meta-view, and general visual rewards, while omitting camera-aware noise warping and trajectory-alignment rewards because they require explicit camera trajectories. VGGRPO (An et al., 2026) is not included because its official implementation has not been publicly released.

Evaluation Metrics. We evaluate generated videos along three dimensions: static consistency, dynamic consistency, and general video quality. For static and dynamic consistency, we pair reward-related and reward-independent metrics to verify gains beyond the optimized objective. General video quality is assessed using third-party VBench metrics (Huang et al., 2024), none of which is used during training.

① *Static Consistency.* We report **Masked Epipolar Error (M.Epi.)** and **Masked Reprojection Error (M.Reproj.)**. M.Epi. is the reward-independent metric: it measures epipolar violations between static-region correspondences. M.Reproj. measures the discrepancy between original and reprojected static regions. Both exclude tracked dynamic subjects. Lower values are better. Because suppressing viewpoint changes can trivially reduce both errors, we interpret them jointly with Dynamic Degree.

② *Dynamic Consistency.* We report **Tracked-Subject DINOv3 Similarity (Subject DINO)** and the **Checklist "Yes" Rate**. Tracked-Subject DINOv3 Similarity (Siméoni et al., 2025) is the reward-independent metric and computes the mean cosine similarity of DINOv3 features in tracked subject regions across adjacent frames. Checklist "Yes" Rate measures the proportion of dynamic-consistency criteria satisfied by each video. Higher values are better. Because a frozen subject can trivially produce high DINO similarity, we interpret it jointly with Dynamicity "Yes" Rate and Dynamic Degree.

③ *General Video Quality.* Following VBench (Huang et al., 2024), we report **Dynamic Degree**, **Motion Smoothness**, and **Aesthetic Quality**, with higher values being better. Dynamic Degree measures the overall amount of motion and helps identify motion suppression, while Motion Smoothness and Aesthetic Quality assess motion continuity and frame-wise visual quality, respectively.

Implementation details are provided in the supplementary material.

4.2 COMPARISON WITH STATE-OF-THE-ART METHODS

We compare WorldAlign with pretrained base models and geometry-aware alignment methods on static consistency, dynamic consistency, and motion preservation. Complementary **qualitative com-**

Table 1: Comparison with base models and state-of-the-art alignment methods on two image-to-video generators. For the Checklist "Yes" Rate, parentheses report the Dynamicity "Yes" Rate over all test videos. M.Reproj. and Checklist "Yes" Rate are reward-related; M.Epi. and Subject DINO are reward-independent. Bold and underlined entries denote the best and second-best results, respectively. Consistency metrics are interpreted jointly with Dynamic Degree because motion suppression can yield deceptively favorable scores, shown in gray. WorldAlign improves both reward-related and reward-independent consistency metrics while preserving overall and subject motion.

	Static Consistency		Dynamic Consistency		VBench		
Method	M.Epi. ↓	M.Reproj. ↓	Subject DINO ↑	Checklist "Yes" Rate ↑	Dynamic Degree↑	Motion Smooth. ↑	Aesthetic Quality ↑
Wan2.2-TI2V-5B							
Base	0.463	0.040	0.922	0.697 (0.63)	0.63	0.957	0.479
Epipolar-DPO	**0.268**	**0.022**	**0.961**	0.646 (0.43)	0.31	0.975	0.481
VideoGPA	0.308	0.024	0.948	0.652 (0.56)	0.54	0.971	0.476
World-R1	0.432	0.038	0.945	0.657 (0.61)	0.56	0.965	0.459
GeoFlow	0.367	0.031	0.934	0.736 (0.74)	0.67	0.971	0.478
WorldAlign (Ours)	0.321	0.026	0.938	**0.781 (0.85)**	**0.72**	**0.979**	**0.488**
Wan2.1-I2V-14B-480P							
Base	0.469	0.037	0.918	0.760 (0.81)	0.62	0.963	0.481
Epipolar-DPO	**0.215**	**0.019**	**0.967**	0.594 (0.53)	0.33	0.976	**0.494**
VideoGPA	0.243	0.022	0.963	0.608 (0.61)	0.42	0.970	0.489
World-R1	0.428	0.034	0.951	0.636 (0.72)	0.53	0.962	0.440
GeoFlow	0.336	0.028	0.936	0.807 (0.83)	0.65	0.975	0.485
WorldAlign (Ours)	0.264	0.024	0.939	**0.856 (0.92)**	**0.74**	**0.978**	0.491

parisons and human evaluation are provided in the supplementary material due to space limitations; both further support the quantitative findings in Tab. 1.

As shown in Tab. 1, the pretrained base models retain moderate motion but exhibit high geometric errors and limited dynamic consistency, confirming that world consistency remains an unreliable capability of pretrained video generators. The alignment methods differ in their ability to improve consistency while preserving motion. Epipolar-DPO and VideoGPA report deceptively low geometric errors, accompanied by substantial decreases in both Dynamic Degree and Dynamicity "Yes" Rate. Their apparent geometric gains are therefore largely attributable to the suppression of camera and subject motion rather than real improvements in world consistency. VideoGPA exhibits less severe motion suppression because it filters candidates with insufficient camera motion during preference construction, whereas Epipolar-DPO has no such mechanism. Our World-R1 reproduction provides only marginal improvements in static consistency while still suppressing subject motion, as reflected by its lower Dynamic Degree and Dynamicity "Yes" Rate. Its reconstruction and meta-view terms evaluate the reconstructed scene through perceptual and semantic similarity without isolating dynamic subjects. In dynamic scenes, subject-motion discrepancies can therefore dominate the reward, diluting the static-consistency signal and biasing optimization toward subject-motion suppression. GeoFlow improves static and dynamic consistency over the base models while increasing both Dynamic Degree and Dynamicity "Yes" Rate. However, its Checklist "Yes" Rate remains 4.5 and 4.9 percentage points below WorldAlign on Wan2.2 and Wan2.1, respectively.

Compared with both the pretrained base models and GeoFlow, WorldAlign achieves lower reward-related M.Reproj. and reward-independent M.Epi. Dynamic improvements similarly hold for the reward-related Checklist "Yes" Rate and reward-independent Subject DINO, while Dynamicity "Yes" Rate and Dynamic Degree also increase. Thus, the gains transfer beyond the optimized rewards without relying on overall or subject motion suppression. Its gains in Motion Smoothness and Aesthetic Quality further indicate the resulting dynamics remain coherent and visually plausible.

4.3 Ablation Study

Tab. 2 evaluates the reward components. Masked reprojection alone reduces geometric errors but suppresses motion; adding the Camera Reward raises Dynamic Degree from 0.42 to 0.70. The

Table 2: Ablation of the proposed WorldAlign on Wan2.2-TI2V-5B. Camera and dynamic rewards are incrementally added to both the direct- and masked-reprojection branches. The results show that masked reprojection avoids imposing a static-scene assumption on the entire video and prevents the suppression of dynamic subject motion. The camera reward alleviates the static collapse observed with masked reprojection alone, while the dynamic reward further improves dynamic consistency.

	Static Consistency		Dynamic Consistency		VBench		
Variant	M.Epi. ↓	M.Reproj. ↓	Subject DINO ↑	Checklist "Yes" Rate ↑	Dynamic Degree ↑	Motion Smooth. ↑	Aesthetic Quality ↑
Base	0.463	0.0396	0.922	0.697 (0.63)	0.63	0.957	0.479
w/ Direct R. Reward	0.298	0.0223	**0.968**	0.628 (0.18)	0.24	**0.992**	0.482
+ Camera Reward	0.324	0.0272	0.963	0.632 (0.22)	0.64	0.982	0.479
+ Dynamic Reward	0.338	0.0291	0.954	0.670 (0.55)	0.69	0.974	0.477
w/ Masked R. Reward	**0.289**	**0.0215**	0.956	0.667 (0.55)	0.42	0.986	0.480
+ Camera Reward	0.317	0.0262	0.927	0.711 (0.64)	0.70	0.968	0.475
+ Dynamic Reward (Full Design)	0.321	0.0260	0.938	**0.781 (0.85)**	**0.72**	0.979	**0.488**

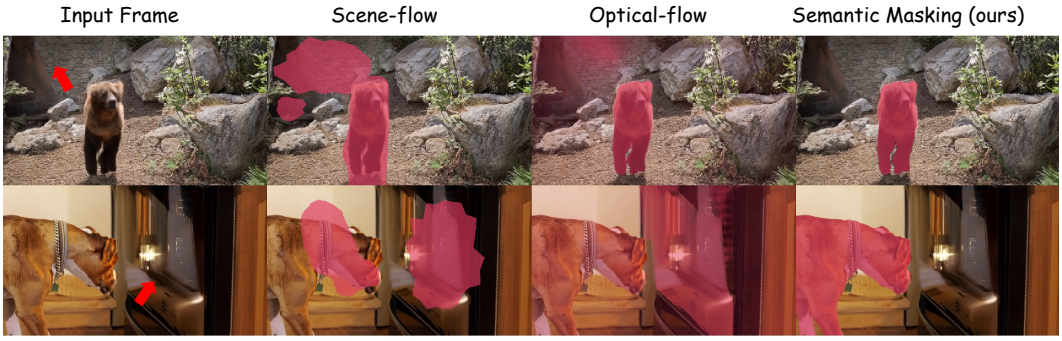


Figure 4: Qualitative comparison of dynamic region identification. The first column shows only the first frame of each video; red arrows mark background regions that distort over time and should remain subject to static consistency evaluation. Red overlays indicate regions identified as dynamic. Scene flow and optical flow, the motion cues used by VGGRPO and GeoFlow, respectively, also mark distorted background regions, whereas our semantic masks more cleanly isolate the subjects.

Dynamic Reward further improves Subject DINO and Checklist "Yes" Rate while retaining the geometric gains. Direct reprojection instead penalizes legitimate subject motion, creating conflicting objectives and yielding poorer static and dynamic consistency than the full masked design. The complete ablation analysis is provided in the supplementary material.

4.4 Comparison of Dynamic Region Identification

Fig. 4 compares dynamic region identification using scene flow (the motion cue used by VG-GRPO (An et al., 2026)), optical flow (the motion cue used by GeoFlow (Ackermann et al., 2026)), and our semantic masking on the same videos. Both motion cues can identify undesired background distortion as dynamic alongside legitimate subject motion, as illustrated by the red overlays at the arrow-marked locations. Filtering these regions out of geometric evaluation or downweighting their depth-consistency contribution weakens feedback on the very background errors that should be corrected. Our semantic masks retain these regions for static evaluation while separating the dynamic subjects, supporting the choice of semantic rather than motion-based region assignment.

5 Conclusion

We introduced WorldAlign, a decoupled 4D reward framework that separates static regions and dynamic subjects and aligns each with a world prior suited to its assumptions. For static regions, a masked reprojection reward leverages a geometry foundation model to assess static consistency without interference from subject motion, while an auxiliary camera-motion reward prevents nearly static solutions. For dynamic subjects, a structured *VLM-as-a-judge* reward uses sample-specific checklists to assess physical motion and appearance consistency. These rewards enable online post-training without architectural changes or human preference annotations. Across two pre-trained image-to-video generators, WorldAlign jointly improves static and dynamic consistency on both reward-related and reward-independent metrics without suppressing overall or subject motion. These results demonstrate the effectiveness of decoupled world-prior alignment for advancing video generation toward more faithful visual world simulation.

AI Use Statement

Generative AI tools were used for manuscript language polishing. The authors verified the cited references, reviewed all AI-assisted edits, and take full responsibility for the final manuscript.

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